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The Editors

Statistical Methods in Medical Research

Dear Editors,

I am pleased to submit the manuscript “**Statistical Evidence for Halachic Vesset Patterns in Natural Menstrual Cycle Data: A Study of Jewish Religious Law and Menstrual Cycle Regularity — A Permutation and Multinomial Sampling Analysis**” for consideration for publication in *Statistical Methods in Medical Research*.

What this paper does. Jewish religious law (*Halacha*) prescribes that women anticipate menstruation based on recurring patterns in their cycle history, collectively known as *vesset*. Despite their centuries-long practical application, no published study has formally tested whether these patterns occur in real menstrual cycle data at rates exceeding chance. This paper fills that gap using three complementary null models — a global permutation null (H_G), a multinomial sampling null (H_{iid}), and a within-woman permutation null (H_W) that conditions on each woman’s observed cycle-length multiset — applied to a publicly available dataset of 1,554 cycles from 118 women (Fehring et al., *Contraception*, 2013).

Methodological contributions. The paper makes three methodological contributions that we believe are of broad interest to readers of this journal:

1. **A joint multivariate test for dependent count vectors.** We introduce a Mahalanobis-distance-based joint test (D_M) that evaluates whether a vector of sequentially computed pattern counts departs from the null distribution, without assuming independence among the counts. This test is applicable to any setting where multiple sequential pattern types are counted in time-series or longitudinal data.
2. **A polynomial hierarchy of pattern types.** We show that the five vesset pattern types form a hierarchy based on the order of the finite-difference operator — degree-0 (constant), degree-1 (linear), degree-2 (quadratic) sequences. We derive three mathematically necessary dependence relationships between the patterns (set-theoretic inclusion, limiting convergence, logical incompatibility) and confirm each with pairwise chi-square tests. This motivates the joint test and explains why Bonferroni correction would be logically inappropriate.

3. **A within-woman permutation null as the primary scientific test.** We introduce a within-woman null (H_W) that permutes each woman's cycle-length sequence within her own record, conditioning on her observed cycle-length multiset. H_W is the most scientifically conservative of the three nulls: it controls for each woman's personal cycle-length distribution and tests only whether patterns arise more often than chance within her own sequence. Under H_W , no pattern is individually significant (all $p \geq .722$) and the joint Mahalanobis test is non-significant ($D_M = 2.95$, $p = .121$). A supplementary multinomial null (H_{iid}) draws each woman's cycle sequence from the fitted empirical distribution; the close agreement of H_G and H_{iid} across all five patterns provides a robustness check on the global permutation results.

Main findings. The primary result is that no vessel pattern is individually or jointly significant under the within-woman null (H_W): all individual $p \geq .722$ and the joint Mahalanobis test yields $D_M = 2.95$, $p = .121$. This shows that the observed pattern counts are consistent with within-woman randomness once each woman's personal cycle-length distribution is controlled for.

Under the global permutation null (H_G), Haflaga departs significantly from chance ($z = 4.51$, $p < .001$; only 2 of 50,000 simulations reaching the observed count). Dilug is nominally significant ($p = .043$) but does not survive Holm correction ($p_{\text{Holm}} = .173$) and should be treated as hypothesis-generating only. A focused 3-pattern Mahalanobis test detects significant multivariate departure from the global null ($D_M = 4.52$, $p < .001$). The contrast between H_G and H_W is itself informative: it shows that Haflaga overrepresentation relative to the population pool reflects partly the composition of individual women's cycle-length distributions, rather than purely within-sequence patterning.

Fit with *Statistical Methods in Medical Research*. This paper sits squarely within the scope of the journal: it develops and applies novel statistical methodology — randomization testing, multivariate distance tests, and null model comparison — to a medical dataset with direct clinical and public-health relevance (menstrual cycle regularity). The interdisciplinary context (Jewish religious law) is unusual, but the statistical methodology and its contributions are entirely general and applicable beyond the halachic setting.

Originality and prior publication. This manuscript has not been submitted elsewhere and is not under review at any other journal. A preliminary analysis appears in the author's PhD dissertation (Ariel University, 2022, in Hebrew), but the present paper extends that work substantially: it introduces the joint multivariate test, the multinomial null model, the polynomial hierarchy with mathematical derivations, the chi-square dependence analysis, and the 5-dimensional geometric visualization, none of which appeared in the dissertation.

Data and code availability. All analysis code and the filtered dataset are publicly available at <https://github.com/dvirross/VesselVectorTest>. The original dataset is archived at Marquette University (https://epublications.marquette.edu/data_nfp/7/).

Suggested reviewers. Given the interdisciplinary nature of this work, I suggest reviewers with expertise in both statistical methodology and reproductive medicine or longitudinal biological data. I am happy to suggest specific names if the editors find this helpful.

I believe this manuscript will be of genuine interest to the readership of *Statistical Methods in Medical Research*, and I look forward to the editorial team's consideration.

Yours sincerely,

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Declarations

Funding

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Conflict of interest

The author declares no conflicts of interest.

Ethics approval

This study is a secondary analysis of a publicly available, fully anonymised dataset (Fehring et al., 2013, archived at Marquette University). No new data collection was conducted. No ethical approval was required.

Patient consent

Not applicable. No new data were collected. The original dataset was collected under the ethical oversight of the original investigators (Fehring et al., 2013).

Data availability

The filtered dataset and all analysis code used in this study are publicly available at <https://github.com/dvirross/VesetVectorTest>. The original NFP dataset is archived at Marquette University: https://epublications.marquette.edu/data_nfp/7/.

Author contributions

Dvir Ross: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing — original draft, Writing — review & editing, Visualization.

Artificial intelligence use disclosure

Portions of the written text in this article were drafted and edited with the assistance of Claude (Anthropic, 2024), a large language model AI assistant. All scientific content, mathematical derivations, statistical analyses, and intellectual contributions are solely the work of the author. The AI tool was used exclusively for language editing, structural organisation of the manuscript, and generation of $\text{\LaTeX}$ and Python code under the author's direct supervision and verification. The author takes full responsibility for the accuracy and integrity of all content presented in this article.

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